



CSE461

Section: 6

Group: 5

Semester: Summer 2026

Robot Project Proposal

Member Details:

1. Name:	Alena Halder Tamo	ID: 22301046
2. Name:	Tahmida Tanjim Talukder	ID: 22101730
3. Name:	Tahsin Raihan Robbani	ID: 22201344
4. Name:	Mofftasim Hossain Sayem	ID: 22299109
5. Name:	Mayen Shahriar Kabir	ID: 23101107

Submission Date: 12 July, 2026

Instructors:

Pratick Roy Chowdhuruy (Lecturer, BRAC University)

Purba Tahia Hassan (Lecturer, BRAC University)

Project Idea 1

1. Project Title: Autonomous Garbage-Collecting Robot with Claw

2. Purpose

Objective: The aim of the project is to design and construct a robot that can move autonomously within a specified area and pick up 1 garbage items from the ground using a mechanical claw, and put it into a bucket mounted on the robot and stop to collect when the bucket is loaded with garbage items. During operation none of this occurs with the intervention of human beings.

Scope: The scope of this robot is indoor or semi-indoor areas like classrooms, labs, corridors or small spaces outdoors such as courtyards. It is provided as a prototype system to show the feasibility of an automated waste collection system at a small scale. The robot works in a flat surface space and can pick up light-weight garbage such as bottle caps, crumpled paper or plastic wrap.

Significance: Garbage collection in public areas is primarily manual and time consuming with sub-optimal hygienic measures. This robot demonstrates how a low-priced, autonomous robot with simple sensors and an Arduino chip can do this without human interaction. These systems could lessen the workload in a university, park, or factory floor, if applied on a large scale. This project illustrates the practical application of one of the most important concepts of robotics: autonomous navigation, sensor fusion and actuator control, for the course.

3. Components

- Microcontroller:

Arduino Uno: The robot's brain. Collects sensor data, executes decision logic and manages all actuator

- Sensors:

Sensor 1 — HC-SR04 Ultrasonic Sensor

Installed on the front of the robot at the middle. When the robot is moving, it is used to detect obstacles. Continuously measures distance ahead of the robot. If an object is found in a specified range, then the robot turns away from it if it is large. This sensor only alerts to whatever is at its height, so a small object such as a bottle on the ground will not set it off.

Sensor 2 — IR Obstacle Sensor Module

Mounted very low on the robot, near claw level, (approx 3-5 cm from the ground). It is used to detect small objects on the ground, which is too small to be detected by the ultrasonic sensor. If this sensor is activated and not the ultrasonic, the robot can recognize that it has

detected garbage and can start the garbage collecting process. The combination of the two sensors enables the robot to distinguish between an obstacle and garbage using only height.

Sensor 3 — Load Cell with HX711 Amplifier Module

Mounted on the waste receptacle on the vessel. Measures the full weight of garbage collected in the bucket. The more items the robot collects, the higher its weight will be. When it reaches a certain threshold (e.g., 200g), the robot stops collecting, sounds an alert buzzer and waits for the bucket to be retrieved.

- Actuators:

Actuator 1 — Servo Motors (x2)

The claw mechanism is controlled by two servo motors. There are two servos, one to open and close the gripper (the actual claw) and the other to raise and lower the arm that carries the claw. Once garbage is found, the arm will drop, the claw will close, the arm will rise and then the claw will let go, dropping the garbage into the garbage bucket above. For this range of motion, these are 180° servo motors.

Actuator 2 — DC Gear Motors via L298N Motor Driver

All movement of the robot is controlled by four DC geared motors (provided with 4WD chassis kit). The L298N Dual H-bridge motor driver module is controlled by the Arduino and is used for controlling motors. This enables the robot to move forward, backward, turn left and turn right. The speed can be controlled by the PWM signals and the robot can be slowed down while approaching a detected object.

- Body/Chassis:

4WD Smart Robot Chassis Kit is an open chassis and is equipped with four DC gear motors, wiring and mounting parts. It offers a sturdy and large platform for mounting the Arduino, sensors, claw arm, bucket and battery.

- Additional Components:

1. L298N Dual H-Bridge Motor Driver Module: Controls DC motors from Arduino
2. HX711 Load Cell Amplifier: Converts load cell analog readings to digital for Arduino
3. Buzzer Module: Sounds an alert when the bucket is full
4. 9V Batteries (x2) with battery holders: Power supply for Arduino and motors
5. Breadboard: For prototyping sensor and component connections
6. Jumper wires: For all electrical connections
7. Small plastic bucket or container: Serves as the onboard garbage bin

8. Screws, zip ties, foam padding: For mounting and securing parts

4. Cost Breakdown (Approx.)

Component	Unit Cost (BDT)	Quantity	Total (BDT)
Arduino Uno R3	990	1	990
HC-SR04 Ultrasonic Sensor	100	1	100
IR Obstacle Sensor Module	55	1	55
Load Cell (HX711 Module)	195	1	195
SG90 Servo Motor (180°)	135	2	270
L298N Motor Driver Module	220	1	220
4WD Chassis Wheels	1,116	1	1,116
DC Motors	95	4	380
Jumper Wires (pack)	120	3	360

Breadboard	160	1	160
9V Battery + Holder	90	2	180
Buzzer Module	20	1	20
Plastic Bucket / Container (for garbage bin)	20	1	20
Miscellaneous (screws, foam, zip ties)	100	1	100
Total Estimated Cost			4166

5. Functionality Breakdown

Functionality 1: Autonomous Navigation and Obstacle Avoidance

In this functionality, the robot is required to navigate and avoid obstacles independently. In this functionality, the robot must autonomously navigate and avoid obstacles.

Overview

The robot is allowed to roam about freely in a specified environment without assistance. In the middle of the road, it continuously looks for obstacles in its way through the front ultrasonic sensor and guides itself around them, without stopping.

Working Procedure

The ultrasonic sensor HC-SR04 is placed in front of the chassis at an approximate height of 15 or 20 cm. It emits ultrasonic waves and measures the time it takes for the echo to return. This check is performed continuously in the main loop of the Arduino. When the distance measured is less

than a value set (e.g. 25 cm), the Arduino tells the L298N motor driver to stop the forward motors and turn left or right to avoid the obstacle. The robot then turns once more, and proceeds forward once again if the path is clear. The sensor is raised in the air, making the logic not trigger when a small object is on the ground, e.g. a bottle cap or crumpled paper, which is below the sensor's beam. It is this separation on the basis of height which enables the robot to distinguish obstacles from garbage without performing any complex image processing or AI.

Functionality 2: Garbage Detection and Auto-Pickup.

Overview

If a small object is detected by the low profile IR obstacle sensor but does not trigger the higher-placed ultrasonic sensor, it is identified as trash and the claw moves automatically to retrieve and place the trash in the onboard bucket.

Working Procedure

The IR obstacle sensor is attached to the robot, low around 3-5cm up from the ground in front of the claw. This sensor keeps scanning for objects as the robot moves forward, at ground level. If it finds any object but the ultrasonic sensor is above it, the Arduino will assume that it is an object on the ground and is small enough to be garbage. Now the Arduino will put the motors into a very slow speed and then stop. The sequence starts with the arm servo lowering the claw towards the object, the gripper servo closing the claw around the object, the arm servo lifting back up and the gripper servo opening the claw over the bucket to drop the object in. The whole sequence runs in servo-angle time mode, and is programmed straight in the Arduino sketch.

Functionality 3: Bucket Capacity Monitoring and Full Alert

Overview

The robot uses a weight to determine the amount of waste the robot has collected by measuring the weight in the garbage bucket. When the bucket is full enough, it will no longer be able to catch any more, and ring an alarm to alert someone to empty it out.

Working Procedure

Garbage bucket is mounted on the robot's chassis and a load cell is placed underneath the bucket. The load cell is connected to the HX711 amplifier module that amplifies the small analog voltage changes from the load cell to a digital signal that the Arduino can read. The Arduino reads the weight measurement from the HX711 each time the robot puts down an object. Once the reading is more than the set value (e.g. 200 grams), the Arduino will terminate the pickup routine completely, and will cause the buzzer to sound continuously. The robot will still be able to move, but will not be trying to lift anything else. This threshold may be adjusted based on testing by

loading the bucket with known weights and observing the readings on the HX711. After the person empties.

Project Idea 2

1. Project Title: Gas Leak and Kitchen Safety Patrol Robot

2. Purpose

Objective: It is designed and constructed as an autonomous robot for a security patrol, which is capable of detecting potential gas leakage, fire danger, and abnormal temperature in a kitchen-like environment. The robot will navigate a small model kitchen, avoid obstacles, senses the environment with a variety of sensors, and warn the users that there is a dangerous condition.

Scope: This Project will be a home, restaurant and kitchen safety monitoring prototype. The robot will automatically patrol a kitchen model to check for the various areas for possible gas leakage, fire and temperature variation. It will be based on the reading that the sensors capture, to decide if an area is safe or dangerous. For the lab demonstration, a small demonstration of a model kitchen (cardboard or foam board) can be built. A general kitchen can be allocated a stove space, gas cylinder space and walk in space in the different sections. The robot will move around the model and avoid obstacles and alert in case of any danger condition. For safety reasons, it may also be demonstrated using values of simulated sensor output or testing methods which have been approved by the faculty.

Significance: In the present world, specifically in Bangladesh accidents occurring from gas leakage and leading to a fatal conditions are too often. Gases leaks and/or kitchen fires can be extremely hazardous in homes, restaurants and small stores. Gas cylinder or gas line is used in many kitchens in Bangladesh, if the gas leakage or abnormally high temperature is detected early, then these accidents can be prevented. In this project a basic robot safety solution is presented that can be used to automatically cover a given area and warn the users in advance to get into trouble. Though this prototype will be exhibited on a small scale model, it illustrates how autonomous robots can be used in the safety of homes and workplaces.

3. Components

- Microcontroller:

Arduino Uno / Arduino Mega: The data will be captured by the sensors and processed by the Arduino, which will then send out the signals to the motors, buzzer, LED, fan and servo motor.

- Sensors:

- Sensor 1 — MQ-2 Gas Sensor:**

- Used to identify conditions of leakage such as LPG, smoke or gas in the kitchen model. If the value of the gas sensor is above a certain value the robot will deem the area unsafe.

- Sensor 2 — Flame Sensor:**

- Used to detect signals of flame/fire danger. This may be demonstrated in the lab using a simulated but safe fire. Or some small flame of lighters can be used too.

- Sensor 3 — DHT11 / DHT22 Temperature and Humidity Sensor:**

- Measures the temperature and humidity level in the kitchen. If the temperature has been high, it could be hazardous.

- Sensor 4 — Ultrasonic Sensor:**

- Detected and used for autonomous movement. It can clear the way of the robot when it is patrolling to avoid the wall, furniture, kitchen objects or model barriers.

- Actuators:

- Actuator 1 — DC Gear Motors:** Used for autonomous robot movement.

- Actuator 2 — Buzzer / LED:** Activated to provide emergency sound and visual alarms in the event of gas leakage, flame hazard, or high temperature.

- Actuator 3 — Mini Fan:** To show ventilation response when gas leakage or lack of air condition is detected.

- Actuator 4 — Servo Motor:** To indicate or stick up a warning flag at the dangerous site.

- Body/Chassis:

1. The chassis of a 2WD/4WD robot car.
2. Wheels
3. Motor mounting frame
4. Platform for Arduino, sensors, battery, and motor driver

- Additional Components:

1. L298N motor driver module
2. Rechargeable battery pack or external battery supply
3. Breadboard
4. Jumper wires
5. Resistors
6. On/off switch

7. Small warning sign
8. Cardboard/foam kitchen model
9. LED indicators for danger status

4. Cost Breakdown (Approx.)

Components	Unit Cost	Quantity	Total
Arduino Uno	990	1	990
Robot Chassis with wheels(2)	1116	1	1116
DC gear motor	95	4	380
L298N motor driver module	220	1	220
MQ-2 gas sensor	290	1	290
Flame sensor	53	1	53
DHT11/DHT22	260	1	260
Ultrasonic sensor	100	1	100
Mini fan	110	1	110
Servo motor	135	1	354
Buzzer,LED	20 + 5	1 + 4	40
Battery	90	2	180
Breadboard,wires	90 + 55	2 + 1 set	235
Cardboard/foam	100	As needed	100
Total Estimated Cost			4428

5. Functionality Breakdown

Functionality 1:Autonomous Kitchen Patrol and Obstacle Avoidance

Overview

The robot will move in an enclosed and small model of a kitchen without humans. It will not be controlled by remote control. The ultrasound sensor will be used to detect the obstacles and help the robot to move in the model kitchen area in a safe manner.

Working Procedure

When the robot is powered on, the motors will start moving, and the Arduino will control the motors via the L298N motor driver. The ultrasonic sensor will be continually measuring the distance from the robot to the close objects. The robot will keep proceeding forward if the path is clear. The Arduino will park the motors and rotate the robot to the left or right when the distance to an obstacle, wall or kitchen object is less than a set distance. The robot can be safely and autonomously moved around in the kitchen model of the house. Power for the robot will be supplied by a battery pack. The sensor(s) and decision making will be handled by the Arduino and the motor driver will handle the movement of the DC motors.

Functionality 2:Gas Leakage and Flame Risk Detection

Overview

The robot will keep track of the gas leak and fire hazard in the kitchen model. The gas sensor MQ-2 will sense gas or smoke-like conditions, the flame sensor will sense the fire-risk signals. If the value given by one of the sensors is higher than the danger level, the robot will identify the area of danger.

Working Procedure

The robot will be navigating through the kitchen model and the MQ-2 gas sensor will be measuring the air around the robot at all times. Once the value obtained from the sensor is greater than the desired gas value, the Arduino will be able to identify if the area is a potential gas-leakage area.

The flame sensor will be monitoring for flame risk signal in the meantime. If the flame sensor receives a bad fire-risk input, Arduino will determine the area as a fire-risk area. Safe simulation also real fire such as small lighter can be used in the laboratory to test the flame sensor for the

flame sensor demonstration. This will keep the project from becoming unsafe but at the same time will show the robot's detection logic.

When gas is detected leaking or there is a risk of fire, the robot will come to a stop close to the area of concern and will be ready to trigger an alarm.

Functionality 3: Temperature Monitoring, Alert, and Ventilation Response

Overview

The robot is designed to measure the temperature and humidity by the use of DHT11 or DHT22 sensor. If the temperature is significantly higher than normal or there is a possibility of gas/flame, the robot will go into its warning system; buzzer, LED, mini fan and a servo controlled warning sign.

Working Procedure

The temperature and humidity data of the kitchen model will be taken by the DHT11/DHT22 sensor. When the temp exceeds a safe value (set by the user), the Arduino will mark the area off as unsafe. The robot will make a final safety decision based on the readings from the gas sensor, flame sensor and the DHT sensor. On detection of a dangerous condition, the robot will stop. The buzzer will turn on to give an audio warning, and the LED will turn on to give a visual warning. In case of gas or inadequate air condition, the mini fan will turn on to show ventilation response. A small warning sign will be raised/pointed to the risky area by the servo motor.

Project Idea 3

1. Project Title: Autonomous Room-Cleaning Robot

2. Purpose

Objective: The idea is to have a self-roaming robot which moves around a room and lowers a mop to address the dirty floor and raises the mop again after the clean, without human intervention.

Scope: It is designed for use in homes, office, classrooms and labs with hard floors. It can clean the hard-to-reach areas that are covered in dust.

Significance: Manual floor cleaning is time consuming and repetitive. A cleaning robot is more efficient because it saves humans their repetitive task. Firstly, it cleans the floor and after cleaning it raises the mop and alerts using a buzzer that it is done with cleaning.

3. Components

1. Microcontroller: Arduino Uno
2. Sensors: ultrasonic sensor (HC-SR04), IR cliff/edge sensor (TCRT5000) of stairs/edge, bump switch for collision.
3. Actuators: 2× DC gear motors (drive), servo motor (lowers/raises the mop), scrub/brush DC motor (cleans), and a buzzer (status).
4. Body/Chassis: 2WD robot chassis kit (2 BO gear motors, wheels, caster wheel, and an 18650 battery holder), underside mount for the microfiber mop pad and the downward facing dust sensor.
5. Additional Components: L298N Motor Driver, 2× 18650 Li-ion Batteries, Breadboard, Jumper Wires, On/Off Switch, and Resistor/Capacitor Circuit for IR LED of the dust sensor.

4. Cost Breakdown (Approx.)

Components	Unit Cost	Quantity	Total
Arduino Uno R3	990	1	990
Ultrasonic sensor (HC-SR04)	100	1	110
IR reflective sensor (TCRT5000)	55	2	110
4WD robot chassis kit	1116	1	1116
L298N motor driver	220	1	220
Servo motor	135	1	135
brush DC motor	50	1	50
Buzzer	20	1	20

18650 Lithium ion battery	180	2	360
Microfiber mop pad	100	1	100
Breadboard	90	1	90
Jumper wires	120	1	120
Total Cost			3,421

5. Functionality Breakdown

Functionality 1: Autonomous Roaming & Obstacle/Cliff Avoidance

Overview:

The robot covers the room by itself while avoiding walls, furnitures.

Working Procedure:

The L298N is used to power the robot, which is controlled by the DC motors. The ultrasonic sensor can detect the presence of obstacles in front. IR cliff sensors are always checking to see if there is a missing floor (stairs), if the sensor detects a missing floor, the robot will go in reverse and pivot away. The ultrasonic has a backup system in the form of bump switches. It is powered by one battery pack, the motor driver separating the motor current from the Arduino logic.

Functionality 2: Movement Control and buzzer alert

Overview:

The robot moves efficiently to better clean the room and gives buzzer to indicate the starting and ending of cleaning work.

Working Procedure:

The robot initiates a buzzer sound when lowering the mop, and after cleaning the floor it gives a buzzer again to notify that cleaning is done. When the robot avoids an obstacle or rotates a turn, it ensures the stable connection between the mop and the floor.

Functionality 3: Mopping Mechanism

Overview:

Robot physically raises and lowers the mop and scrubs the floor.

Working Procedure:

Once it activates “mop on”, the servo lowers the microfiber mop pad down to the floor and the brush motor spins or vibrates over the dirty area to clean. the drive motors roll slowly over the dirty spot. When the arduino calls “mop off” (dust is below threshold), the motor power is turned off and the servo lifts up the pad from the floor to prevent dragging/streaking from room to room.